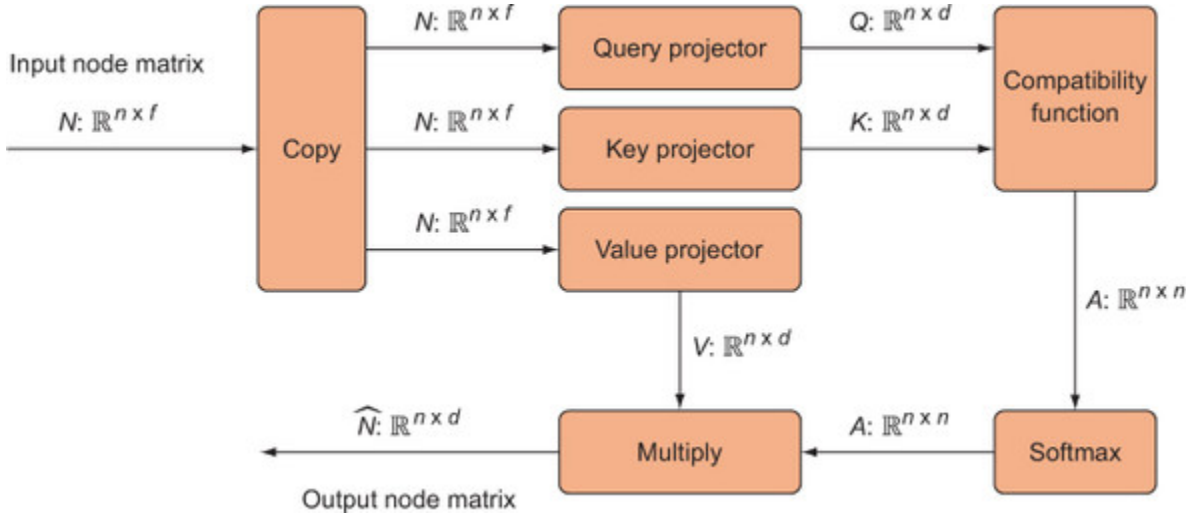
Figure 10.13. A high-level view of a self-attention-based relational module. The input to a relational module is a node matrix N: ..n×f with n nodes each with an f-dimensional feature vector. The relational module then copies this matrix for a total of three copies, and projects each one into a new matrix via a simple linear layer without an activation function, creating separate query, key, and value matrices. The query and key matrices are input to a compatibility function, which is any function that computes how compatible (similar in some way) each node is to each other node, resulting in a set of unnormalized attention weights, $A:\mathbb{R}^{n\times n}$. This matrix is then normalized via the softmax function across the rows, such that each row's values will sum to 1. The value matrix and normalized attention matrix are then multiplied, $\hat{N}=AV$. The output of the relational module is then usually passed through one or more linear layers (not depicted).



Let's take a single pair of nodes to make this concrete. Say we have a node (which is a feature vector), a:$\mathbb{R}^{10}$, and another node, b:$\mathbb{R}^{10}$. To calculate the self-attention of these two nodes, we first will project these nodes into a new space by multiplying by some projection matrices, i.e., $a_Q = a^TQ$, $a_K = a^TK$, $a_V = a^TV$, where the superscript T indicates transposition, such that the node vector is now a column vector, e.g., a^T:$\mathbb{R}^{1\times 10}$, and the corresponding matrix is Q:$\mathbb{R}^{10\times d}$, such that $a_Q= a^TQ$:$\mathbb{R}^d$. We now have three new versions of a that may be of some different dimensionality from the input, e.g., a_Q,a_K,a_V:$\mathbb{R}^{20}$. We do the same for the node b. We can calculate how related a is to itself first by multiplying (via the inner product) its query and key together. Remember, we compute *all* pairwise interactions between nodes, including self-interactions. Unsurprisingly, objects are likely to be related to themselves, although not necessarily, since the corresponding queries and keys (after projection) may be different.